

BACKLOG BUDDY ? STUDY MATERIAL

VLSI Design ? Syllabus & Reference Books

Subject Code: EC502 Semester: 5

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Unit-wise syllabus, recommended reference books and question paper pattern.

Unit 1: CMOS Fabrication and Layout

- Definition, key concepts and significance of CMOS Fabrication and Layout
- Important diagrams / derivations / algorithms for CMOS Fabrication and Layout
- Solved examples and previous-year questions on CMOS Fabrication and Layout

Unit 2: CMOS Inverter and Logic Gates

- Definition, key concepts and significance of CMOS Inverter and Logic Gates
- Important diagrams / derivations / algorithms for CMOS Inverter and Logic Gates
- Solved examples and previous-year questions on CMOS Inverter and Logic Gates

Unit 3: Sequential Circuit Design

- Definition, key concepts and significance of Sequential Circuit Design
- Important diagrams / derivations / algorithms for Sequential Circuit Design
- Solved examples and previous-year questions on Sequential Circuit Design

Unit 4: Design for Testability

- Definition, key concepts and significance of Design for Testability
- Important diagrams / derivations / algorithms for Design for Testability
- Solved examples and previous-year questions on Design for Testability

Unit 5: VLSI Design Flow

- Definition, key concepts and significance of VLSI Design Flow
- Important diagrams / derivations / algorithms for VLSI Design Flow
- Solved examples and previous-year questions on VLSI Design Flow

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.